Cover Letter

A Lightweight Detection Pipeline for Earthquake Early Warning and Its Evaluation on Three Seismic Networks

H. Oğuz Bolat,Sibel Çimen,Bülent Bolat,Tülay Yıldırım

Dear Editors-in-Chief,

Please find enclosed the manuscript “A Lightweight Detection Pipeline for Earthquake Early Warning and Its Evaluation on Three Seismic Networks”, which we submit for exclusive consideration as a research article in Computers & Geosciences. We confirm that the submission follows the journal’s requirements and includes every item of its submission checklist.

The manuscript addresses both of the journal’s scopes. Computationally, it presents ayzek, a self-contained pipeline for the detection stage of earthquake early warning that also reports a P-wave displacement magnitude. Its hand-written inference reproduces the reference implementations to within floating-point rounding, its replays of archived data are deterministic, and the same build runs on a Raspberry Pi 4. It also presents a replay protocol, implemented as a reproducible workflow, that measures detectors at their operating point with chance correction, paired tests and catalog-quiet data, whose settings, including the baselines’, were fixed in a public, timestamped release before any test waveform was fetched. Geoscientifically, it measures how a learned detector, two classical triggers and two published models behave on continuous records of three networks under that protocol: the Greek networks served by the National Observatory of Athens and New Zealand’s GeoNet, which the learned detector had not seen, and the Turkish KO network the detector was trained on, held out in time. At the settings matched on unmatched alarms the learned detector detected 22.4 more chance-corrected events than STA/LTA of 177 on the Greek sequences, though with twice its unmatched alarms, and 18.1 more of 151 on the New Zealand sequences, with equal unmatched alarms; the KO test, under its target in its two-month pool, was not significant. Causal PhaseNet was equivalent to the learned detector on both unseen networks, so the gain is that of learned detectors over an amplitude trigger rather than of ours. On the catalog-quiet windows the learned detector’s alarm rate was not distinguishable from STA/LTA’s on New Zealand or Greece, where an exploratory analysis traces half of its alarms to one station’s emergent noise that STA/LTA ignores, and its alarms came 1.2 to 2.9 s later. On a 728-day record at one station, two detectors with nearly equal ROC-AUC differed by a factor of 1.6 in recall at one false alarm per day.

The source code, model weights, the protocol and the test settings, and the reproduction scripts are in a public repository under the GNU General Public License, archived on Zenodo, with the details listed in the section “Code availability”. The test waveforms are open at their data centers and are not redistributed; the released request records and file hashes let anyone fetch and check the same data. The AFAD development waveforms cannot be redistributed, and public KOERI datasets reproduce the development results.

The use of generative AI tools during this work is declared in the manuscript, and the authors reviewed and edited all content and take full responsibility for it. The manuscript has not been published and is not under consideration elsewhere, all authors approve its submission, and they declare no competing interests.

Sincerely,

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Highlights

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- Hand-written neural networks reproduce the trained models to within rounding
- Ensemble scores a window in 5.2 ms; a Raspberry Pi 4 serves 52 stations
- 18–22 more events than STA/LTA on unseen networks; 2× unmatched alarms on Greece
- Quiet-data alarm rates not distinguishable from STA/LTA's; alarms 1.2–2.9 s later
- Causal PhaseNet equivalent to the learned detector within ±10% on unseen networks

A Lightweight Detection Pipeline for Earthquake Early Warning and Its Evaluation on Three Seismic Networks

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ABSTRACT

Early-warning detectors should be able to run continuously on modest hardware, yet learned ones usually need a machine-learning runtime and are evaluated on curated windows or tuned on their test records. We present ayzek, a lightweight detection pipeline whose hand-written networks reproduce the trained models to within rounding, and a replay protocol, fixed before any test data were fetched, for measuring detectors at their operating point with chance correction, paired tests and catalog-quiet data. The pipeline's three 142 059-parameter networks score a window in 5.2 ms, about 1% of the 0.5 s scoring interval, so one desktop thread serves 95 stations and a Raspberry Pi 4 serves 52 with identical alarms. All settings, the baselines' included, were chosen on Turkish development data before testing on the Greek networks and New Zealand's GeoNet, unseen by the learned detector, and the KOERI network, held out in time. At its development unmatched-alarm level, the learned detector detected 150.3 of 177 Greek events after chance correction against 127.9 for a short-term average/long-term average (STA/LTA) trigger (difference 22.4, 95% interval 11.7–33.5), though with 371 unmatched alarms against 189, and 84.2 of 151 New Zealand events against 66.1 (18.1, 7.2–29.5); KOERI's 36 events gave 2.5 (−0.5 to 5.9). Causal PhaseNet was equivalent within ±10% on both unseen networks, so the gain is not specific to our detector. Its catalog-quiet alarm rates were not distinguishable from STA/LTA's (New Zealand 0.9 against 1.6 per day; Greece 19.3 against 20.7, half from one noisy station), and its alarms came 1.2–2.9 s later. Code and protocol are open.

1. Introduction

Earthquake early warning (EEW) systems detect an earthquake from the first seconds of P waves and alert sites before the strong shaking reaches them. Public systems share a design: a telemetered network, a pipeline that detects and characterizes the event, and an alert policy. Japan's nationwide system has run since 2007 on point-source estimates from the stations nearest the hypocenter (Hoshiba et al., 2008), joined in 2018 by PLUM, which propagates observed shaking instead of a source (Kodera et al., 2020). Taiwan's system reports an event within 20 s from a strong-motion network (Hsiao et al., 2009) densified by low-cost MEMS sensors (Wu et al., 2013). ShakeAlert's EPIC and FinDer were tested on a replay platform before public rollout (Kohler et al., 2018; Cochran et al., 2018; Chung et al., 2019; Kohler et al., 2020); Mexico's SASMEX has alerted for thirty years (Suárez, 2022); Israel's TRUAA (Nof and Kurzon, 2021) and Istanbul's threshold-based system (Alcik et al., 2009) are regional examples. In each, a detector runs continuously on every station, and the P travel time to the stations needed to declare an event bounds the warning the system can give (Allen et al., 2009).

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In Europe, PRESTo (Zollo et al., 2009) is a free regional platform, tested on Greek Ionian stations (Bracale et al., 2021), and the Virtual Seismologist and FinDer run as modules of the SeisComP monitoring framework (Behr et al., 2016; Massin et al., 2021; Porras et al., 2021); EPIC has been replayed on Turkish data (Tunç et al., 2024; Marcou et al., 2026; Budakoğlu et al., 2026). Meier et al. (2020) ran PLUM, EPIC and FinDer on 219 Japanese earthquakes with common scoring, the closest precedent to the replay comparisons of this study. Low-cost designs move processing toward the sensor, from fixed smartphones with a cloud back end (Brooks et al., 2021) to PLUM on Raspberry Shake nodes (Prasanna et al., 2022); deep-learning workflows such as QuakeFlow (Zhu et al., 2023) and single-station models such as CREIME (Chakraborty et al., 2022) run under Python runtimes, offline or in the cloud. All of these detect with amplitude or picker triggers, or with learned models that need a runtime, inside a hosting framework.

Classical triggers such as the ratio of short- to long-term averages (Allen, 1978; Withers et al., 1998) and multi-band pickers such as FilterPicker (Lomax et al., 2012) are inexpensive and time an onset to the sample, but an amplitude criterion sensitive enough for small earthquakes also fires on impulsive cultural and instrumental transients. Deep networks trained to detect earthquakes (Perol et al., 2018; Ross et al., 2018; Mousavi et al., 2019b) and to pick phases (Zhu and Beroza, 2019; Mousavi et al., 2020) are mostly evaluated on curated windows with balanced classes, as the standardized datasets (Mousavi et al., 2019a) and model collections (Woollam et al., 2022) encourage, although learned pickers do run on continuous data (Mousavi et al., 2020; Yeck et al., 2021) and transfer well between regions at similar distances (Münchmeyer et al., 2022).

Curated windows hide the cost that matters in operation: over hours without seismicity, a detector's false alarms accumulate. On balanced windows, a metric such as the area under the receiver operating characteristic curve (ROC-AUC) says little about the false-alarm rate at the operating point. ROC-AUC is insensitive to class prevalence (Fawcett, 2006), which precision–recall analysis (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015) and partial areas at low false positive rates (McClish, 1989) address; in seismic detection the operating region is extreme, one window false alarm per day among 172 800 decisions per station and day being a false positive rate below 10^{-5}. When a detector is evaluated on continuous records, it and its baselines are usually tuned on records from the same network and period. Comparisons of detectors at a fixed alarm budget on continuous records remain rare, and rarer still with false alarms measured on catalog-quiet data.

In this study we evaluate detectors on continuous recordings at a fixed alarm budget. We built ayzek, a detection pipeline whose station stage is a learned detector trained on Turkish data, and chose every setting of it and of four baselines, an STA/LTA trigger, FilterPicker, PhaseNet and EQTransformer, on development replays. The test protocol was fixed and timestamped before any test waveform was fetched; the detectors were then replayed on earthquake sequences and catalog-quiet windows from Greece, New Zealand and, held out in time, the Turkish KO network.

The computational contribution of this study is twofold. The first is ayzek, a lightweight, self-contained pipeline

from miniSEED decoding to the network alarm and a first magnitude, with hand-written inference and no machine-learning runtime, verified against its PyTorch models, deterministic in replay, and running unchanged on a Raspberry Pi 4. Unlike the systems above, its station stage is a learned detector in one binary that runs on a single-board computer, without a hosting framework. The second is a replay protocol, implemented as a reproducible workflow, that measures detection at the operating point. Chance matches with the catalog are corrected by shifting the catalog in time, only the stations that raised an alarm are matched, detectors are compared with paired tests, and false alarms are counted on data without cataloged seismicity.

The geoscientific contribution is a measurement of how learned and classical detectors behave on continuous seismic records from three networks under that protocol: the five Greek networks served by the National Observatory of Athens and the New Zealand national network, neither used to train the learned detector, and the Turkish KO network on which the detector was trained, held out in time. The pipeline's learned detector is compared with a recursive STA/LTA trigger, FilterPicker (Lomax et al., 2012) and the published EQTransformer (Mousavi et al., 2020) and PhaseNet (Zhu and Beroza, 2019) models, to test whether any gain over the classical trigger is specific to our detector. In this study an alarm is a network detection with a first magnitude rather than a warning that an operational system could issue, and all results come from replays of archived data. The study asks whether the advantage of a learned detector over classical triggers, measured on development data, survives on networks it has not seen, at equal false-alarm rates.

2. Material and methods

2.1. Development and test data

All choices in this study were made on development data; the test results come from data that none of these choices used, with one exception, the per-station STA/LTA ratios of Section 2.6. The learned detector was trained on 6 s three-component windows from 181 stations of the KO (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 1971), 6G (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 2013b) and IJ (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 2013a) networks of the Kandilli Observatory and Earthquake Research Institute (KOERI), around cataloged events of $M \geq 2$ up to 29 August 2026, partitioned by station into training, validation and test sets (Section S19). The development replays (Table 1, Figure 1) come from the continuous archive of broadband stations of the Turkish National Seismic Network (TU; Disaster and Emergency Management Authority (AFAD), 1990), operated by the Disaster and Emergency Management Authority (AFAD) in the Marmara and northern Aegean regions and absent from the training set. With three public KOERI sets, they comprise two tuning windows, two sequences and 23 quiet windows (102.8 h). One KOERI set overlaps the detector's training data and serves as a demonstration only. Events are scored only at stations where they are visible, with the signal-to-noise ratio of the 2–20 Hz vertical component around the IASP91 P

arrival t_P (times in seconds),

$$\mathrm{SNR} = \mathrm{RMS}\big(x[t_P - 1,\ t_P + 12]\big)/\mathrm{RMS}\big(x[t_P - 60,\ t_P - 10]\big), \tag{1}$$

of at least 3.

Table 1
Development replays. All settings of this study were chosen on them, and no test result comes from them.

Dataset	Network	Start (UTC)	Duration	Stations	Catalog events	Role
Sındırgı	TU	2025-11-10 18:04:04	30 min	3	9	tuning
Marmara Sea	TU	2025-04-23 08:29:01	3 h	8	57	tuning
Sındırgı M_w 6.1	TU	2025-08-10 16:00	3 h	6	66	tuning (baselines)
Simav M_w 4.8	TU	2026-04-11 14:00	3 h	6	56	tuning (baselines)
Quiet windows (23)	TU	2024-05-22 to 2025-03-07	102.8 h	7–8	0	alarm rate
Quiet window	KO	2024-01-03 11:25:00	5.8 h	6	0	alarm rate, public
Sındırgı M_w 4.9	KO	2025-11-10 18:04:00	31 min	6	9	tuning (baselines), public
Marmara M_w 6.2	KO	2025-04-23 09:00:00	1 h 45 min	8	37	demonstration, public

Note. Catalog events are counted within 250 km of the station centroid. The learned detector's trigger rule was chosen on the first two replays; the baselines' final settings on the four event sequences and the KO M_w 4.9 window (Section 2.6). The two AFAD sequences and the quiet windows also served, before the release date, as this study's development test bed (Section S20). Stations are listed in Table S1; of 15 KO stations, 11 contributed training or validation windows.

The tests use three networks, each reported separately and not pooled (Table 2, Figure 2): the Greek networks HL (National Observatory of Athens, Institute of Geodynamics, Athens, 1975), HT (Aristotle University of Thessaloniki, 1981), HA (University of Athens, 2008), HP (University of Patras, 2000) and HC (Technological Educational Institute of Crete, 2006), served by one EIDA node; the New Zealand network NZ (GNS Science, 2021); and KO (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 1971), the network the detector was trained on, held out in time from 30 August 2026, after the last training window of any model. Of the Greek and New Zealand data, only four 6 h windows per network from before the test period were used, to set the per-station STA/LTA ratios.

A station is used if it records three HH channels at 100 Hz with at least 90% coverage of the test period and its components can be rotated to north and east; coordinates and gains come from each network's StationXML (Table 2).

Sequences. Each network's test pool holds all events of $M \geq 4.5$ in its catalog within its test period that have at least five usable stations within 150 km; the catalogs are EMSC's (European-Mediterranean Seismological Centre (EMSC), 2026) for KO, NOA's for Greece and GeoNet's (GNS Science, 1970) for New Zealand. In time order, each such event starts a sequence unless it lies within 3 h after the first event of an earlier sequence with which its stations share at least

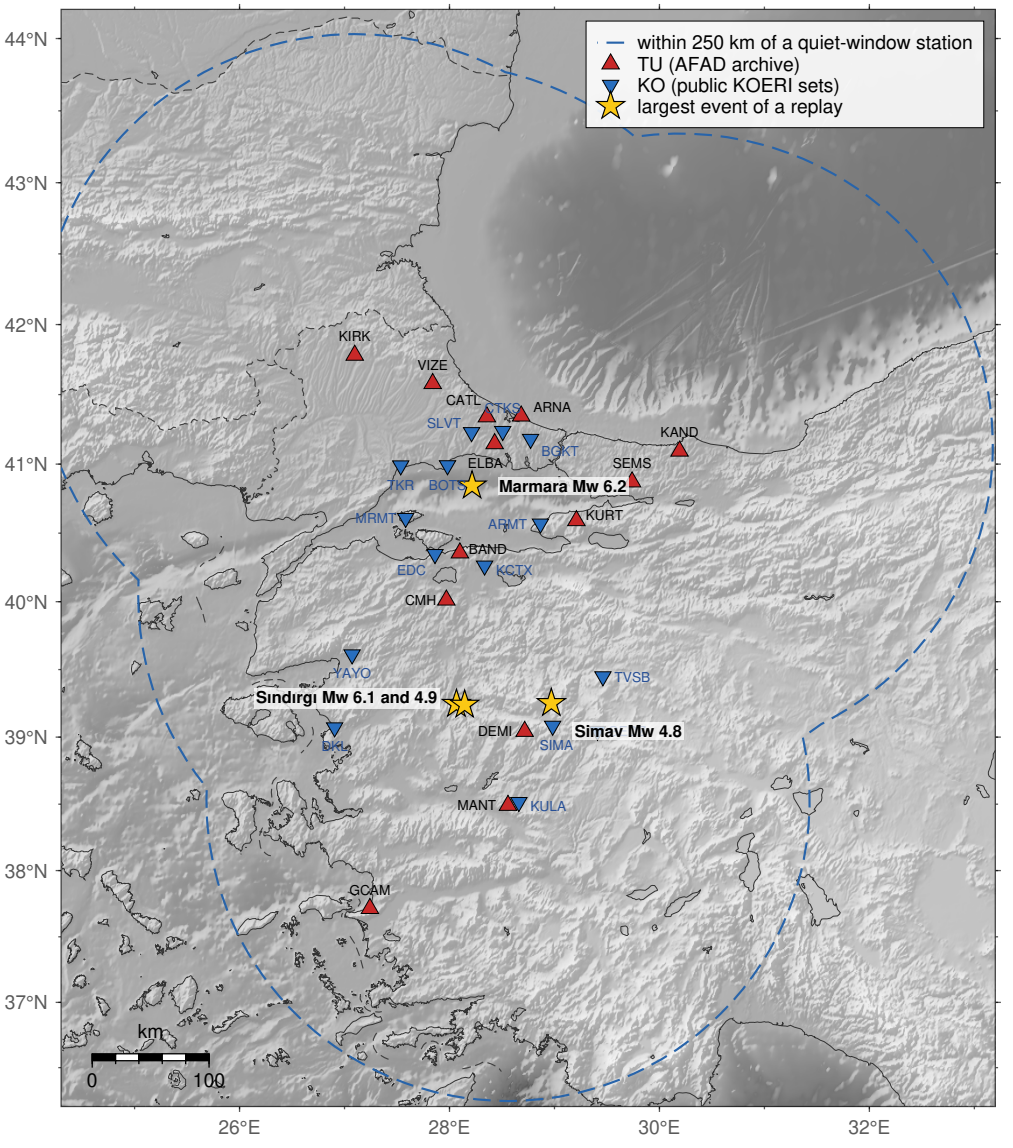


Figure 1: The development data: TU stations of the AFAD archive, KO stations of the public KOERI sets, the largest event of each replayed sequence, and the area within 250 km of the quiet-window stations, in which no cataloged event may lie during a quiet window. Relief from ETOPO 2022 (NOAA National Centers for Environmental Information, 2022).

one; it then belongs to that sequence. A sequence is replayed for 3 h from 10 min before its first event on its nearest usable stations within 150 km, at most eight and at least five, and its cataloged events within 250 km of the stations' centroid are scored as on the development data. The test periods end at the release date (2026-10-30 12:55:36 UTC). Each sequence is assigned, in time order, to this study or to a reserve for later tests by a pseudo-random draw (seed 42). This study's sequences are taken in time order until their scorable cataloged events, counted from the catalog before any waveform is fetched, reach the target, 240 on KO and 150 on Greece and New Zealand, the crossing sequence included whole; a network that does not reach its target uses all of its sequences and is reported as under the target, as

Table 2
Test networks, each reported separately.

Group	Networks (operator)	Data center	Catalog	Period from	Target	Stations usable / probed
Greece	HL (NOA), HT (AUTH), HA (University of Athens), HP (University of Patras), HC (Crete)	NOA EIDA node	NOA	2023-01-01	150	81 / 200
New Zealand	NZ (GeoNet)	GeoNet	GeoNet	2023-01-01	150	36 / 134
KO	KO (KOERI)	KOERI EIDA node	EMSC	2026-08-30	240	123 / 155

Note. All test periods end at the release date, 2026-10-30 12:55:36 UTC. Target: scorable cataloged events of this study's sequences (Section 2.6). Stations: usable over the test period (three HH channels at 100 Hz, at least 90% coverage, rotatable components) out of those probed, determined from the availability listings when the sequences were selected, after the release date. For New Zealand the listing is the day files of GeoNet's open-data bucket, which bound coverage from above, since GeoNet runs no availability service. NOA, National Observatory of Athens; AUTH, Aristotle University of Thessaloniki; HC is operated by the Technological Educational Institute of Crete. All test data postdate the training data of every model. STEAD (Mousavi et al., 2019a), on which the phase picker and the original EQTransformer weights were trained, holds earlier records (2003–2018) of 3 of the 57 Greek test stations (HA.ACOR, HL.ATH, HP.AMT) and of no New Zealand or KO test station.

KO's two-month pool was expected to be.

Quiet windows. Each network also contributes up to 24 windows of 6 h, drawn at random over its test period with seed 43 on the station sets of its sequences, and accepted only if no cataloged event lies within 250 km of any of their stations during the window or 30 min on either side and their data cover 90% of it, small distant events being disregarded if 10 000 draws give fewer than 24. The first 30 min of each window are not scored.

2.2. Detectors

The pipeline's own detector, called the learned detector throughout, scores 6 s windows. Each window (600×3 samples at 100 Hz) is detrended, tapered, band-passed at 1–45 Hz, standardized per component and compressed with an inverse hyperbolic sine. The filter is zero-phase but applied within the window, whose last sample is the current time, so that no later sample is used; the standardization uses the window's own statistics, which removes gain and site amplitude. Three strided convolutional layers reduce the window to 75 steps of 96 channels, and a bidirectional LSTM with 48 units per direction, a four-head self-attention block with residual connection (Vaswani et al., 2017) and a small perceptron produce a single logit (142 059 parameters). The arrangement is that of CRED (Mousavi et al., 2019b) with the self-attention of EQTransformer's encoder (Mousavi et al., 2020), adapted from 1D2D-EDL (Wang and Zhao, 2025). The network was trained with label smoothing, three seeds are averaged, and the ensemble reaches a ROC-AUC of 0.9785 on the station-disjoint curated test set. A P/S picker, run only after a trigger, provides picks for location (Section S19).

The recursive STA/LTA band-passes the trace with fourth-order Butterworth sections, forms the squared vertical component or the three-component energy, and agrees with SciPy (Virtanen et al., 2020) and ObsPy (Beyreuther et al.,

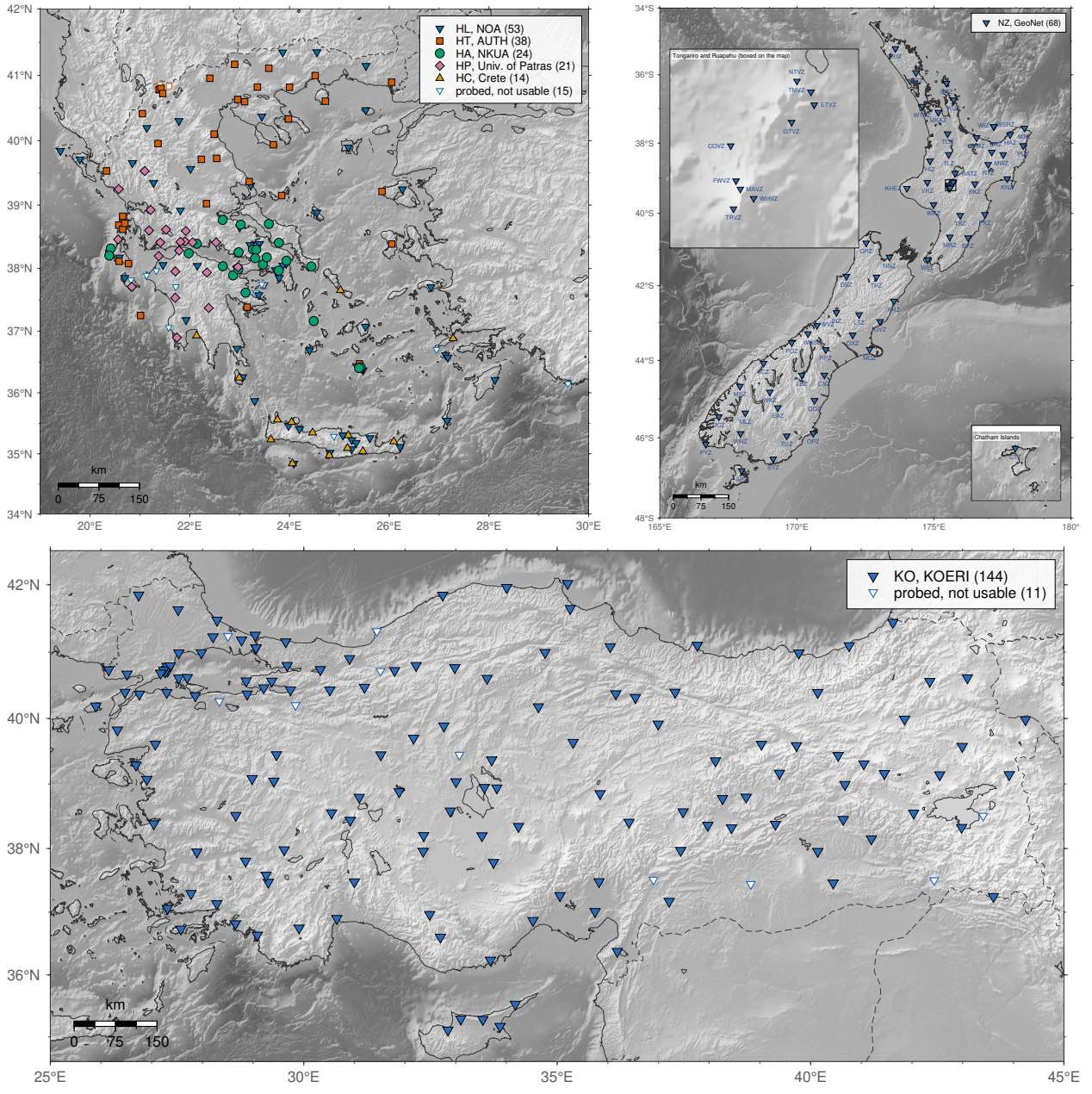


Figure 2: Stations of the test networks: Greece (top left; HL, HT, HA, HP and HC by symbol), New Zealand (top right; the Tongariro and Ruapehu stations in a zoom inset, and the Chatham Islands station, east of 180°, in a second inset) and KO (bottom). Filled symbols are the stations usable in at least one of the four pre-test windows that set the per-station STA/LTA ratios, open symbols in their network's shape the stations probed there but not usable. The maps were drawn before the release date and show no event; the sequences selected afterwards are listed in Table 4. Relief from ETOPO 2022 (NOAA National Centers for Environmental Information, 2022).

2010; Krischer et al., 2015) to within rounding (Section S2). FilterPicker takes the maximum over a bank of band-pass characteristic functions and declares a pick when its mean over a window reaches a second threshold; the pipeline's port of Lomax's C code gives identical picks (Section S12). EQTransformer and PhaseNet run through SeisBench

0.12.6 in place of the learned detector, each 6 s window receiving the largest model output within it, EQTransformer's detection trace or PhaseNet's P probability. In the *causal* mode the model input ends at the end of the window, as in real-time operation; in the *look-ahead* mode it is centered on it, an offline upper bound (Section S8). Each baseline replaces only the learned detector's output; triggers are limited to one per 15 s per station, and everything downstream is unchanged.

2.3. Real-time pipeline

The pipeline, written in C++23, runs from ingestion to location and magnitude (Figure 3). Each station is served by an ingestion and a processing thread that feed one network stage through an ordered message bus, and the build is one binary and a directory of weights. Records are released in the arrival order of a live feed, and a ring buffer whose positions give the sample times neither drops nor blocks writes (Section S2).

Windows are scored on an absolute grid every 0.5 s. Label smoothing makes the learned detector's outputs saturate between 0.90 and 0.907, so that a single threshold at $p \geq 0.9$ depends on the third decimal. The adopted rule, chosen on the tuning replays (Table S18), fires when eight consecutive windows reach $p \geq 0.8$, 3.5 s after the first, or when one window reaches $p \geq 0.9$; a trigger re-arms after two windows below 0.3, and no window within 15 s of the previous trigger may start a run. The P time of a learned trigger is the first sample in its window at which a recursive STA/LTA (2–20 Hz, 1 and 30 s, vertical) running alongside reaches a ratio of 3, otherwise 3.5 s into the window; this sample always lies in the past, and the anchoring adds no latency. Two detections are associated when their P times differ by no more than the P travel time between the stations plus 3 s, an event is declared when two stations agree, and later detections at a station within 40 s are absorbed as its coda. Location, a two-stage grid search in a half-space from accepted P and S picks, serves the report after the alarm and is not evaluated (Section S16).

All layers are implemented by hand and load the PyTorch (Paszke et al., 2019) parameters unchanged; agreement tests on real waveforms put the worst-case error of the detector logit at 6.0×10^{-7}, and repeated and concurrent replays produce byte-identical reports. On a Ryzen 5 5600 a window of the three-model ensemble costs 5.2 ms against 0.0005 ms for the STA/LTA, and one thread sustains 95 stations; on a Raspberry Pi 4 the static aarch64 build raises the same alarms at the same times and four cores sustain 52 stations (Table S4). The tests run the release's static x86-64 build, whose reports on all development replays equal the development build's.

2.4. Magnitude and early location

The magnitude is estimated from the peak displacement of the P wave, Pd (Wu and Kanamori, 2005; Kanamori, 2005), rather than learned. At each station with a gain, the raw vertical passes through a causal chain: a two-pole high-pass at 0.075 Hz, integration and the same high-pass. Pd(τ) is the largest absolute displacement in the τ s after the station's onset, for τ of 1, 2, 3, 4, 5, 7 and 10 s, with the noise level taken over the 10 s ending 1 s before it. The

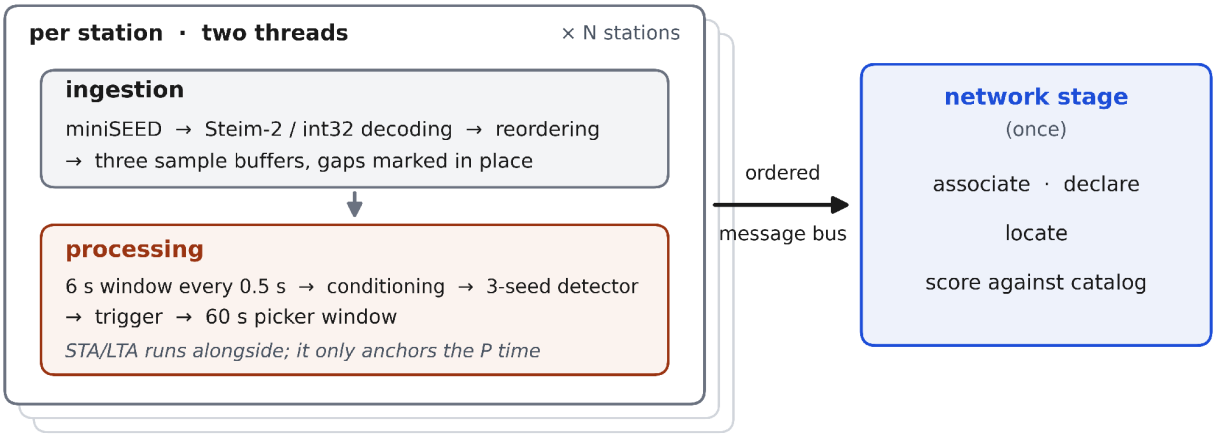


Figure 3: Architecture of the real-time pipeline. Each station runs an ingestion and a processing thread; a network stage associates station detections, declares, locates and sizes events and scores them against a catalog. The recursive STA/LTA never triggers; it only supplies the reported P time of a learned trigger.

relation between $\log_{10}$Pd, magnitude and hypocentral distance has two magnitude slopes and three distance slopes per window and a term per station. An event's magnitude is the posterior mean of this relation's likelihood over its stations' values under a Gutenberg–Richter prior; a value below three times its noise level enters as an upper bound. The relation was fitted on 96 908 values from 11 806 KO events in a separate station-disjoint data set; on its test stations, from M3 upward, the estimate at $\tau = 5$ s has a bias within ± 0.05 and a mean absolute error of 0.12 to 0.21. With each estimate the pipeline reports the posterior probability of three size classes, below M4, M4 to 5.5, and above; for events larger than about M6.5 the estimate and its class are lower bounds. On KO the calibration's gains, station terms and noise levels apply, with its exclusion of channels whose noise lies far from their median. On Greece and New Zealand the gains come from each set's StationXML, and a vertical channel is accepted only if its response divided by the stated sensitivity lies within 0.8–1.25 between 0.1 and 5 Hz. On these two networks the station terms are set to zero and no noise levels apply, so that no channel is excluded for being dead or in a coda. KO's station terms average +0.07 to +0.10 in $\log_{10}$Pd, so zero terms bias an average station by about +0.1, against the relation's scatter of about 0.3.

Pd requires hypocentral distances before the picks of the 60 s picker window can locate an event. Each declared event is therefore also located from its detections' onsets alone, by a grid search in a uniform half-space at 10 km depth from three stations on, the onsets coming from a FilterPicker that runs beside the learned detector and triggers nothing (median onset error 0.06 s on 418 development arrivals). This location serves the Pd distances only, once its 68% radius is below 20 km, until a pick location replaces it (Section S16). None of this touches detection, association or matching: all development replays give the same alarms with and without it.

2.5. Scoring

Alarms without a catalog counterpart are called unmatched rather than false, because the catalogs are incomplete for small events during aftershock sequences; the false-alarm rate is measured on the quiet windows, as alarms per day with exact Poisson intervals. A declared event matches a catalog event if its epicenter lies within 30 km and its origin within 5 s, or if the predicted P arrival at a station that raised the alarm lies within 6 s of that station's P time, P being predicted in a uniform half-space. Each alarm is credited to the one catalog event it fits best, and each catalog event to the first alarm credited with it, so that no alarm counts for two events and a second alarm for the same event counts as unmatched. The chance level c is the detection rate with all catalog times shifted by ± 127, ± 263 and ± 391 s, as the time slides of gravitational-wave analysis measure accidental coincidences (Was et al., 2010), and with observed rate o the chance-corrected rate is

$$\hat{o} = (o - c)/(1 - c), \tag{2}$$

reported as an equivalent number of events; inside aftershock sequences most shifted events land within 60 s of a real one, so c overstates chance matching and the correction is conservative (Section S17). Window probabilities are cached, which reproduces full inference exactly. Two detectors are compared on per-event outcomes by an exact McNemar test and a paired bootstrap of the difference in chance-corrected detections (10 000 resamples within each sequence).

2.6. Test protocol

The protocol was committed to the pipeline's public repository and timestamped on Zenodo before any test waveform was fetched (release v1.0.0, commit 3e43064, 2026-10-30, doi:10.5281/zenodo.23806743), with the release that is tested, the hashes of each model file and of the static binary, and the code that fetches, replays and scores the data. It is summarized here and given in full in Section S24.

Detectors. The learned detector keeps the rule fixed on the tuning data. All baselines, STA/LTA, FilterPicker and the published PhaseNet and EQTransformer models, were retuned on 29 October 2026 on all development event replays, the four AFAD sequences and the KOERI M_w 4.9 window, 197 cataloged events, by three rules. The first takes the most chance-corrected detections at no more unmatched alarms than the learned detector, the second the fewest unmatched alarms at no fewer chance-corrected detections, and the third the most chance-corrected detections at no more alarms than the learned detector on the AFAD quiet windows. Because some unmatched alarms in aftershock sequences are uncataloged earthquakes, more of them for a sensitive detector, the third rule matches the rate on data without cataloged seismicity instead. For the published models the third rule is applied within the weights the first

Table 3
Detector settings of the tests, selected on the development event replays. Rule 1 matches the learned detector's unmatched alarms, rule 2 its detections and rule 3 its alarms on the quiet windows.

Detector	Rule	Setting	Corr.	Unm.	Quiet
Learned	–	0.8 × 8, or 0.9 once	128.2	52	16
STA/LTA	1	1–10 Hz, 0.5/10 s, ratio 8, vertical	110.3	47	63
	2	1–10 Hz, 1/10 s, ratio 4, vertical	131.6	113	224
	3	1–10 Hz, 0.5/10 s, ratio 10, 3 components	64.8	27	12
STA/LTA, per station	–	rule 1, each station's own ratio (3–15)			
FilterPicker	1	6/10/0.4 s, thresholds 20 and 10	102.5	44	291
	2	6/10/0.4 s, thresholds 15 and 8	130.9	101	954
	3	6/10/0.4 s, thresholds 6 and 15	52.0	9	9
PhaseNet, causal	1	INSTANCE, 0.8 × 1	119.7	52	49
	2	INSTANCE, 0.5 × 4	128.3	70	29
	3	INSTANCE, 0.7 × 2	126.1	62	15
EQTransformer, causal	1	INSTANCE, 0.7 × 8	62.1	49	7
	2	INSTANCE, 0.3 × 8	128.5	183	26
	3	INSTANCE, 0.6 × 2	94.4	130	14
PhaseNet, look-ahead	1	INSTANCE, 0.7 × 4	123.2	52	12
	2	INSTANCE, 0.6 × 8	135.9	61	14
	3	INSTANCE, 0.7 × 1	138.1	74	16
EQTransformer, look-ahead	1	INSTANCE, 0.6 × 4	113.0	50	11
	2	original, 0.7 × 8	140.0	88	13
	3	original, 0.6 × 2	153.2	116	16

Note. Rules: 1, equal unmatched alarms; 2, equal chance-corrected detections; 3, equal quiet alarms. Corr. and Unm.: chance-corrected detections and unmatched alarms over the five development event replays (197 cataloged events); Quiet: alarms on the 23 AFAD quiet windows (102.8 h). STA/LTA: band, STA and LTA lengths, trigger ratio, components; it re-arms at a ratio of 1.5. FilterPicker: its three windows and two thresholds. Learned detector and published models: probability threshold × consecutive windows, with their weights (Michelini et al., 2021); the instant branch is off for the published models. The candidates were the original, STEAD, INSTANCE and ETHZ weights of EQTransformer and the STEAD, INSTANCE and ETHZ weights of PhaseNet. The look-ahead variants use input past the window and run offline only. The per-station STA/LTA has no development counts; its ratios come from each test network's own pre-test windows.

two select. A per-station STA/LTA stands for the per-station tuning of operational practice: the first rule's setting with each station's own trigger ratio, the lowest of eight values at which the station's trigger rate on four 6 h windows from before the test period does not exceed its network's median rate at ratio 8 (Section S24). Table 3 lists all settings. The pipeline's transformer detector is fixed in the same release and tested under the same protocol at its defaults; it is reported in a companion paper.

Endpoints. The primary endpoint, on each network, is (i) the difference in chance-corrected detections between the learned detector and STA/LTA at equal unmatched alarms over its sequences, with a 95% paired bootstrap interval resampled within each sequence. The secondary endpoints are (ii) the same difference against STA/LTA at equal quiet alarms and (iii) against each other detector; (iv) equivalence with causal PhaseNet by two one-sided tests at 5% with a margin of 10% of the scorable events; (v) each detector's alarm rate on the quiet windows, also without the

network's two stations of most STA/LTA detections; (vi) the rule's adjacent settings (thresholds 0.75 and 0.85, 6 and 10 windows); (vii) the share of alarm detections declared before the predicted S, and the primary comparison on the events detected before S; (viii) on KO, the primary comparison on test-split stations apart from training-split ones; (ix) the magnitude's error, bias and class accuracy by band and the share of alarmed events with an estimate, for events no deeper than 40 km; and (x) alarm delay and warning time before S. Because the matcher predicts P and S in a uniform half-space, 3 to 12 s late for events deeper than 100 km, each detection endpoint is also reported for the events no deeper than 40 km. No multiplicity correction is applied, each secondary result is labeled as such, and nothing is retuned on test data. The stopping rule counts events from the catalog before any waveform is fetched; the endpoints use the events ayzek scores (those with a depth and a magnitude, with origin from 30 s after the replay's start to 30 s before its end), a few fewer per sequence.

Prospective tests. The same protocol is applied to New Zealand and KO data recorded after the release date, data that did not exist when the method was fixed: all qualifying sequences from that date on, until the target or 12 months, each fetched once when four weeks old, with 24 quiet windows drawn once the span has closed (Section S24). The archive delivers these data intact, so the test replays them and is not a live run.

3. Results

Section 3.1 summarizes the development results, which are not a test. Sections 3.2 to 3.6 report the protocol's endpoints on the three test networks, each separately. On a continuous 728-day record at one development station, the learned detector and the pipeline's STA/LTA have nearly equal event ROC-AUC (0.9892 and 0.9910) but the learned detector finds 1.6 times as many events at one window false alarm per day, and four offline detectors rank in the reverse order of event AUC at that budget (Section S14).

3.1. Development results

On the development data, at the settings in use before the retuning, the learned detector detected 36.4 of 66 events after correction for chance against 21.7 for STA/LTA at comparable unmatched alarms on the Sındırgı M_w 6.1 sequence (difference 14.7, 95% interval 5.9–23.7) and 35.1 of 56 against 29.1 on Simav, not significant; causal PhaseNet matched it on both. On the 102.8 h of quiet windows it raised 3.74 alarms per day, STA/LTA 75.4 at its earlier setting and 14.7 at the retuned rule-1 setting that was tested (Table 3). Section S20 gives these results in full.

3.2. The selected test sets

The selections were run after the release date, on 30 and 31 October 2026, and each set was fetched and verified against its request record (Table 4). Greece reached its target with 11 sequences and 178 scorable events, New Zealand

with 13 sequences and 154. KO's two-month pool held three qualifying sequences, two of them assigned to this study, with 36 scorable events against a target of 240; both are used, as the protocol provides. KO's quiet draws also fell short, 14 windows against 24. Two Greek stations, ZKR and KOKK, delivered no data in their window, so two Greek replays run on 5 and 7 stations. Four of New Zealand's 13 sequences start with an event deeper than 170 km.

Table 4
The test sets selected after the release date.

Network	Catalog events	Qualifying sequences	This study's / selected	Scorable events (target)	Stations per seq.	Quiet windows (draws)	Quiet hours
Greece	90 382	84	42 / 11	178 (150)	6–8	24 (2 059)	132
New Zealand	81 990	29	16 / 13	154 (150)	5–8	24 (551)	132
KO	5 693	3	2 / 2	36 (240), under	7, 6	14 (20 000, relaxed)	77

Note. Catalog events: all events of the network's catalog in the test period; KO's 36 are under its target. Qualifying: sequences by the rule of Section 2.1; this study's are those the draw assigned to it, selected those taken before the target was reached. Scorable events: the stopping rule's count. Quiet windows: accepted of the draws made; KO's first 10 000 draws accepted 6, the relaxed pass 14. Scored quiet hours: 5.5 h per window after the unscored lead-in. The sequences, their stations and the windows are listed in Section S21.

3.3. Detection: the primary endpoint

Table 5 gives each detector on each network. On Greece, the learned detector detected 150.3 chance-corrected events of 177 against 127.9 for STA/LTA at equal unmatched alarms: a difference of +22.4 (95% interval +11.7 to +33.5; exact McNemar $p = 0.0001$, 25 events detected by the learned detector only against 4 by STA/LTA only). The two settings did not, however, raise equal unmatched alarms on Greece, as they had on the development data: the learned detector raised 371 and STA/LTA 189, two thirds of the learned detector's in two sequences (121 at Epirus and 104 at Evia; Section 4). Restricted to the 175 events no deeper than 40 km the difference is the same, +22.4 (+11.3 to +33.6).

On New Zealand, the learned detector detected 84.2 chance-corrected events of 151 with 20 unmatched alarms, against 66.1 for STA/LTA at equal unmatched alarms, also with 20: a difference of +18.1 (95% interval +7.2 to +29.5; exact McNemar $p = 0.002$, 25 events detected by the learned detector only against 7 by STA/LTA only). Restricted to the 137 events no deeper than 40 km (14 deeper, none without a depth), the difference is +17.9 (+7.2 to +29.0).

On KO, under the target with 36 scorable events, the learned detector detected 15.5 chance-corrected events with 9 unmatched alarms against 13.0 for STA/LTA with 8: a difference of +2.5 (95% interval −0.5 to +5.9; $p = 0.25$, 3 events detected by the learned detector only and none by STA/LTA only). All 36 events are shallower than 40 km, so the restricted result is the same. Most of KO's events are below M2 in EMSC's catalog, where neither detector finds much, and both detected all four events of M4.5 and above.

Outside the protocol, resampling 10- or 15-min blocks of origin time instead of single events, as on the development data, widens the intervals to at most +10.1 to +35.6 on Greece and +6.7 to +30.0 on New Zealand, and leaving out any one sequence keeps both above zero. New Zealand's largest sequence holds 89 of its 151 events; without it the

difference is +7.4 (+3.2 to +12.3) of 62, the same share (Section S26).

Table 5
Each detector at its setting of Table 3 on the three test networks: chance-corrected detections (Corr.), unmatched alarms (Unm.), the difference from the learned detector with its 95% paired bootstrap interval, and alarms per day on the quiet windows.

	Greece (177)				New Zealand (151)				KO (36), under target			
Detector, rule	Corr.	Unm.	Difference	Quiet	Corr.	Unm.	Difference	Quiet	Corr.	Unm.	Difference	Quiet
learned	150.3	371	–	19.3	84.2	20	–	0.9	15.5	9	–	9.7
STA/LTA, 1	127.9	189	+22.4 (+11.7, +33.5)	20.7	66.1	20	+18.1 (+7.2, +29.5)	1.6	13.0	8	+2.5 (−0.5, +5.9)	20.9
STA/LTA, 2	138.9	401	+11.4 (+1.9, +21.0)	103.8	80.8	49	+3.4 (−7.2, +13.9)	6.0	16.6	56	−1.1 (−6.9, +4.6)	111.9
STA/LTA, 3	100.3	66	+50.0 (+36.7, +63.5)	3.3	40.7	7	+43.5 (+32.0, +55.1)	0.4	8.0	1	+7.5 (+3.0, +12.7)	3.4
STA/LTA, per station	130.4	160	+19.9 (+9.7, +30.5)	4.2	54.4	15	+29.8 (+17.8, +41.6)	2.2	10.0	4	+5.5 (+1.4, +9.9)	2.5
FilterPicker, 1	128.4	193	+21.9 (+11.0, +32.9)	44.9	89.2	67	−5.0 (−16.0, +5.6)	6.2	11.0	9	+4.5 (+0.6, +8.8)	31.2
FilterPicker, 2	133.7	581	+16.5 (+5.1, +28.1)	270.2	105.0	247	−20.9 (−32.6, −9.5)	28.0	13.7	56	+1.8 (−2.6, +6.4)	201.7
FilterPicker, 3	90.7	25	+59.6 (+46.9, +72.3)	2.4	58.1	7	+26.1 (+15.0, +37.2)	0.9	8.9	1	+6.7 (+2.3, +11.7)	1.6
PhaseNet causal, 1	145.3	229	+5.0 (−2.1, +12.9)	12.9	78.4	15	+5.8 (−3.4, +15.4)	0.4	10.0	3	+5.5 (+1.4, +9.9)	7.2
PhaseNet causal, 2	144.5	357	+5.8 (−2.3, +14.6)	25.4	98.7	43	−14.5 (−24.8, −4.7)	1.4	15.8	11	−0.3 (−3.3, +2.7)	11.5
PhaseNet causal, 3	143.9	271	+6.4 (−1.6, +14.6)	12.9	87.5	26	−3.3 (−12.5, +5.9)	0.4	11.9	6	+3.6 (+0.4, +7.7)	6.9
EQTransformer causal, 1	89.9	219	+60.4 (+45.7, +75.3)	3.6	25.9	13	+58.3 (+46.8, +70.0)	0.0	5.0	1	+10.5 (+5.2, +16.1)	1.6
EQTransformer causal, 2	137.3	687	+13.0 (+2.9, +23.5)	16.6	65.5	50	+18.6 (+7.8, +29.4)	1.3	10.8	10	+4.8 (−0.1, +9.9)	11.5
EQTransformer causal, 3	128.0	470	+22.2 (+10.6, +34.6)	8.6	44.4	27	+39.8 (+28.3, +51.1)	0.6	9.9	3	+5.7 (+0.6, +10.9)	4.4
Look-ahead, offline only												
PhaseNet, 1	143.9	266	+6.3 (−3.3, +16.0)	7.1	85.6	20	−1.5 (−11.9, +9.1)	0.2	11.0	3	+4.5 (+0.6, +8.8)	4.7
PhaseNet, 2	144.1	301	+6.2 (−2.9, +15.3)	8.2	88.7	25	−4.5 (−15.1, +6.3)	0.7	12.0	6	+3.5 (+0.3, +7.6)	6.9
PhaseNet, 3	145.6	352	+4.6 (−5.0, +14.6)	12.0	94.8	31	−10.6 (−21.3, −0.4)	0.7	12.0	7	+3.5 (−0.7, +7.9)	7.5
EQTransformer, 1	115.5	291	+34.8 (+22.7, +47.0)	7.1	30.5	9	+53.7 (+40.9, +66.3)	0.2	1.0	2	+14.5 (+8.6, +20.6)	3.4
EQTransformer, 2	150.6	291	−0.4 (−9.9, +9.0)	6.4	97.4	51	−13.3 (−24.8, −1.9)	2.5	17.8	3	−2.2 (−5.4, +0.0)	6.2
EQTransformer, 3	154.7	385	−4.4 (−13.5, +4.7)	9.3	104.0	73	−19.9 (−30.9, −8.9)	3.3	18.7	4	−3.2 (−6.7, −0.2)	9.0

Note. Rules as in Table 3; the per-station STA/LTA uses rule 1's setting with each station's own ratio. In parentheses after each network, the events ayzek scored. Unm.: unmatched alarms over the network's sequences. Quiet: alarms per day on the network's quiet windows (Table 4); the counts and the rate ratios with their exact intervals are in Table S22. The same table for the events no deeper than 40 km is Table S23. Rule 1 against the learned detector is the primary endpoint; all other rows are secondary.

3.4. Detection: secondary endpoints

All results of this section are secondary; Section S24 gives each in full, network by network.

The learned detector stayed ahead of the baselines whose settings kept their development operating point (Table 5): of STA/LTA at equal quiet alarms by +50.0 events on Greece and +43.5 on New Zealand, of the per-station STA/LTA by +19.9 and +29.8, and of causal EQTransformer by +60.4 and +58.3. FilterPicker at equal unmatched alarms was 21.9 events behind it on Greece and level on New Zealand, there at 67 unmatched alarms against 20. Causal PhaseNet was equivalent to the learned detector on Greece (+5.0; 90% interval −0.9 to +11.6, margin ±17.7) and New Zealand (+5.8; −1.8 to +13.9, ±15.1), with fewer unmatched alarms on both, and behind it on KO (+5.5, +1.9 to +9.5, ±3.6); the margins are nearly as wide as the primary differences.

On the quiet windows the learned detector's rate did not differ distinguishably from that of STA/LTA at equal unmatched alarms on New Zealand (0.9 against 1.6 per day, 5 alarms against 9; rate ratio 0.56, 0.15 to 1.85) or Greece (19.3 against 20.7; 0.93, 0.71 to 1.22), and was lower on KO (9.7 against 20.9; 0.46, 0.29 to 0.72); on Greece and KO the classical settings matched on quiet alarms raised 1.6 to 3.4 per day and the published models' 4.4 to 12.9. Leaving out each network's two stations with the most STA/LTA detections lowers the Greek rates to 12.2 and 5.8 (rate ratio 2.09, 1.35 to 3.30), which reverses their order, and changes no other ordering. The rule's neighbors with 6 windows

detected 11 to 20 more events than the rule on New Zealand and none more than 0.2 more on Greece; those with 10 windows detected fewer on both. At least 97% of the learned detector's alarm detections preceded the predicted S on each network, and restricted to the events detected before S the primary difference is +15.7 on both unseen networks. KO's station-split comparison is empty, all its stations being training-split stations. Of the events of M4.5 and above, the learned detector missed one, a Greek MLh 4.9 that came 19 s after an M1.9 both detectors alarmed.

3.5. Magnitude

On Greece, under StationXML gains and zero station terms, all 10 alarmed events of M4 and above received an estimate. For the seven MLh 4 to 5.5 events no deeper than 40 km the final estimate has a bias of +0.09 and a mean absolute error of 0.11, with the size class right for all seven at the first and the final estimate; the MLh 5.8 of 29 March 2024 received M5.8 6.5 s after its alarm, and 56 of the 143 smaller alarmed events received an estimate, biased by +0.13. On New Zealand, under the same conditions, all 10 alarmed events of M4 and above received an estimate and 8 of the 77 smaller ones did. An estimate needs a station whose Pd exceeds three times its noise; the smaller Greek events that received one had a station a median 12 km away, those that did not 41 km, and New Zealand's smaller events 42 km. For the eight MLv 4 to 5.5 events no deeper than 40 km the final estimate has a bias of +0.03 and a mean absolute error of 0.20, with the size class right at the first and the final estimate for all eight; the expected bias of zero terms, about +0.1, is not visible at this count. On KO, with its calibration, three of the four events of M4.5 and above received estimates within 0.4 of their catalog magnitudes, and the M5.0 none. Deeper events are in Section S22.

3.6. Latency

Table 6 gives the alarm delays and warning times. The learned detector's alarms came a median 1.2 s (Greece), 1.8 s (New Zealand) and 2.9 s (KO) after STA/LTA's. Warning time uses the S arrival the pipeline picked where it has one and the predicted one otherwise. The release's analysis tool read only the predicted arrivals, a bug that dropped the stations with a pick, those nearest the source, so that its nearest station ahead of S lay farther away (on New Zealand a median 105 km instead of 78 km) and its warnings were longer; as the protocol's deviation rule requires, the table gives both (Section S24).

4. Discussion

We argue that a detector for early warning should be evaluated at the operating point at which it will be used, on data it has not seen, and under a comparison fixed before those data were chosen, and this study was designed accordingly; at one development station, event-level ROC-AUC barely separated two detectors that the operating point separated clearly (Section S14). The held-out tests answer the question posed in the Introduction. On New Zealand, a network the learned detector had not seen, the advantage held: 18 more chance-corrected events than STA/LTA at

Table 6
Alarm delay after origin and warning before S at the nearest station still ahead of S, for the learned detector and STA/LTA at rule 1.

Network	Detector	Alarm delay (s), median (IQR)	Events with a station ahead of S	Median warning (s)	
				Fixed	Release's tool
Greece	learned	12.0 (10.0–13.5)	151 / 153	4.5	4.8
	STA/LTA	10.8 (9.0–12.4)	132 / 132	5.3	5.1
New Zealand	learned	16.0 (14.5–18.5)	87 / 87	7.7	13.0
	STA/LTA	14.2 (13.5–17.0)	69 / 69	8.5	14.7
KO	learned	20.8 (17.5–23.5)	16 / 16	3.7	5.4
	STA/LTA	17.9 (13.1–21.5)	13 / 13	7.6	7.7

Note. Events: detected events with a station ahead of S at the alarm, of those detected. Fixed: S picked where the pipeline has a pick, predicted otherwise; release's tool: predicted S only, counting 126 of 130 and 103 of 104 Greek events (Table S24). KO's M_w 5.3 was declared 21.5 s after origin by the learned detector and 18.9 s by STA/LTA, 10.2 and 12.9 s before S at the nearest station (16.0 and 18.6 s by the release's tool).

equal unmatched alarms, out of 151, with an interval well clear of zero, and the same within 2.5 events for the shallow events alone and for the events detected before S. On Greece the difference was larger, 22 events out of 177, and the same for the shallow events, but it came with twice STA/LTA's unmatched alarms, so the comparison was not at an equal operating point there. On KO, with 36 events, the difference had the same sign and an interval including zero, as the protocol anticipated; it is evidence of consistency, not a test.

On these data the classical detectors are neither obsolete nor superior. On development data the learned detector gained in M2–4 events that an amplitude criterion cannot separate from noise, and it kept that sensitivity at a quarter of the quiet-data alarm rate of STA/LTA at rule 1 (3.74 against 14.7 per day), though STA/LTA at rule 3 was quieter still. On New Zealand its rate stayed low, 0.9 alarms per day, but at 5 alarms against STA/LTA's 9 not distinguishably lower. On Greece and KO it did not stay low: 19.3 and 9.7 per day, while the classical baselines matched on quiet alarms stayed near their development rates. An analysis outside the protocol, made after the Greek result, separates the causes (Section S25). In the two Greek sequences with the most unmatched alarms, the learned detector's 225 divide into 77 that locate within 30 km of the mainshock, 7 that follow within 60 s events missing from NOA's catalog, 135 that could not be located and 6 located elsewhere. Causal PhaseNet, at the same development operating point and with equivalent detections, raised 115 there, 53 located in the sequences: of the learned detector's excess, 24 locate in the sequences and 84 cannot be located, at Evia mostly with one station. On the quiet windows, one 6 h window in Crete holds 51 of the learned detector's 106 Greek alarms, nearly all with station STIA, on emergent signals that no catalog explains; STA/LTA did not fire on them, so the protocol's exclusion of the two stations noisiest for STA/LTA did not remove them. Without that window the learned detector raised 10.4 alarms per day against 21.2, with that exclusion 12.2 against 5.8. The learned detector thus finds a few more small uncataloged earthquakes than the other detectors, but most of its excess alarms on Greece are station responses that STA/LTA and PhaseNet do not share, and

an unmatched or quiet-window alarm count does not separate the two. STA/LTA alarmed earlier at the median on every network, by 1.2 s on Greece, 1.8 s on New Zealand and 2.9 s on KO, since a window detector can trigger only once the P wave occupies part of its 6 s window; for early warning this is a real cost. Classical triggers also cost orders of magnitude less and time the onset to a sample, which is what the pipeline uses them for. Causal PhaseNet, trained on Italian data, was equivalent to the learned detector on both unseen networks, with fewer alarms on Greece. The gain over STA/LTA is therefore that of learned detectors generally, not of this one; what this study adds is a pipeline that runs such a detector, with 142 059 parameters per ensemble member and no machine-learning runtime, and a protocol that measures it.

Much of the protocol rests on the computational design: cached window probabilities and deterministic replays made trigger grids of hundreds of settings and the retuning of all baselines practical, and the release, the request records and the file hashes let anyone fetch the same test data again and check them. Only complete replays exposed two failures of the data path, among them an earlier version that dropped samples under load and lost 15 of 44 detections (Section S2).

4.1. Limitations

The KO test is far under its target, with 36 events in two sequences and 14 quiet windows, so its interval is wide; the prospective KO test will add to it. The catalogs are incomplete for small events, so that unmatched alarms only bound false alarms, and EMSC's catalog lists fewer small events in KO's region than AFAD's does, which lowers KO's scorable counts and raises all detectors' unmatched alarms there. The matcher predicts P and S in a uniform half-space, so events deeper than about 150 km are scored as missed for all detectors alike, which the depth-restricted results address. Parameters shared by all detectors, such as the 40 s coda absorption and the 15 s between a station's triggers, were not swept. The learned detector responds to emergent station noise that STA/LTA ignores (Section 4), which the protocol's station exclusion, keyed to STA/LTA, does not catch. The ranking on a continuous record rests on one station, and the published models ran with pre-trained weights that fine-tuning might improve. The input is a replay of archived files; a live SeedLink client, with its latency, packet loss and clock problems, was not part of the tests.

5. Conclusions

A self-contained, verified real-time pipeline and a replay protocol with chance correction, paired tests and catalog-quiet data measure seismic detectors at their operating point. Under this protocol, on the Greek and New Zealand networks that the learned detector had not seen, it found 22.4 (Greece) and 18.1 (New Zealand) more chance-corrected events than STA/LTA at the settings matched on development data, with 95% intervals 11.7–33.5 and 7.2–29.5. On

Greece this came at twice STA/LTA's unmatched alarms, on both networks at quiet-data alarm rates not distinguishable from STA/LTA's, though on Greece the comparison turns on which stations are included, and with alarms 1.2 to 1.8 s later; on KO, held out in time and under its target, the difference was 2.5, with an interval including zero. Causal PhaseNet was equivalent to the learned detector on both unseen networks, so the gain is that of learned detectors over an amplitude trigger. At one development station, two detectors with nearly equal event AUC differed by a factor of 1.6 in recall at one window false alarm per day. Our results indicate that detectors for early warning are better compared on continuous records at fixed alarm budgets.

CRediT authorship contribution statement

H. Oğuz Bolat: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Sibel Çimen:** Writing – review and editing. **Bülent Bolat:** Writing – review and editing, Supervision. **Tülay Yıldırım:** Writing – review and editing, Supervision.

Code availability

Name of the software: ayzek. Developer: H. Oğuz Bolat (oguz.bolat@std.yildiz.edu.tr), first available in September 2026. Program language: C++23, with Python tools for model export and evaluation. Hardware requirements: an x86-64 or aarch64 processor; tested on a desktop computer and on a Raspberry Pi 4. Software required: Meson and Ninja and a C++23 compiler (GCC 16 or Clang); no machine-learning runtime. Program size: about 9 400 lines of C++ and 6.5 MB of model weights. License: GNU General Public License, version 3 or later. The source code, model weights, the protocol, the test settings, the documentation and the scripts that reproduce each result (`tools/reproduce_all.sh`, with expected values in `bench/claims.json`; `tools/run_heldout.py` for the tests) are available at `https://github.com/hogib/ayzek`. The tests run release v1.0.0 (commit 3e43064, 30 October 2026), archived at `https://doi.org/10.5281/zenodo.23806743`; the release carries the SHA-256 of its static x86-64 binary and of each model file, and the tool that fetches the test data at commit 440a2d9, as a tarball; the tool's repository is `https://github.com/hogib/ayzek-heldout`. The FilterPicker port is derived from the C code of Lomax et al. (2012) (GNU GPL version 2 or later). Software versions are listed in Table S9.

Data availability

The AFAD waveforms are available from AFAD's data portal (TDVMS, `https://tdvms.afad.gov.tr`) on registration and request, and cannot be redistributed. The KOERI (Kandilli Observatory and Earthquake Research Institute (KOERI), Boğaziçi University, 1971), Greek (National Observatory of Athens, Institute of Geodynamics, Athens, 1975; Aristotle University of Thessaloniki, 1981; University of Athens, 2008; University of Patras, 2000; Techno-

logical Educational Institute of Crete, 2006) and New Zealand (GNS Science, 2021) waveforms are open, KOERI's under citation of the network, the Greek networks' under CC BY 4.0 from the NOA EIDA node, and GeoNet's under CC BY 3.0 NZ. The test waveforms are not redistributed: they are fetched from these data centers by the tool released with the pipeline, which keeps for each set a record of the request (`dataset.json`: services, stations, channels and windows, the canary check, and what arrived) and the SHA-256 of each file it wrote, so that a later fetch can be checked against the data used in this study. The selection of the sequences and quiet windows, the station tables and the catalog requests are released with it. After the release the fetch tool received three commits (658ae03, 5d088c5 and 6fda3e2) that change only how it retries and splits its requests to NOA's availability service, none of what it selects or writes; KO was selected and fetched at 440a2d9, New Zealand and Greece fetched at 6fda3e2, as each set's record states. The catalogs are AFAD's for the development replays, EMSC's (European-Mediterranean Seismological Centre (EMSC), 2026) for the KO tests, which the European-Mediterranean Seismological Centre serves through its FDSN event service with the Turkish agencies' solutions (KOERI runs no event service, and AFAD's answered each request made when the protocol was written with an empty list), NOA's FDSN event service for Greece and GeoNet's (GNS Science, 1970) for New Zealand. The three public KOERI development sets are attached to the pipeline's release. The offline evaluation code is available at `https://github.com/SismoKaos-Research/cnn_earthquake`, the training code of the detector at `https://github.com/hogib/cascade` and that of the picker at `https://github.com/hogib/sphase`. Maps use ETOPO 2022 (NOAA National Centers for Environmental Information, 2022), GSHHG (Wessel and Smith, 1996) and GMT (Wessel et al., 2019).

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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the University of Athens, the University of Patras and the Technological Educational Institute of Crete. The KO test catalog was retrieved from the FDSN event service of the European-Mediterranean Seismological Centre.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Opus 5, Claude Opus 5.5 and Claude Fable 5.1 (Anthropic) to develop and refactor parts of the pipeline code and to edit the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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